PRG4206 MACHINE LEARNING

Project Execution & Evaluation Outputs

A Comparative and Explainable Machine Learning System for AI-Generated Text Detection

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Dataset: HC3 — Human ChatGPT Comparison Corpus
Models: Logistic Regression, Linear SVM, XGBoost

Important status note: This copy is a summary of the saved benchmark artefacts already included in the project. It is **not** evidence that Google Colab has been run. Open and execute *notebooks/00_PRG4206_Colab_Run_and_Outputs.ipynb* to run the checks in Colab; the notebook will regenerate this PDF with the actual run record.

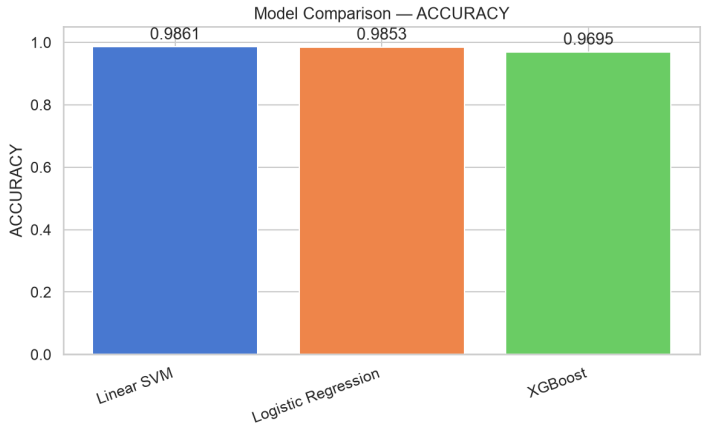
1. Model comparison on the held-out test set

The three saved classifiers were evaluated on the same 11,900-example test partition. Values below are read from the bundled verified evaluation file.

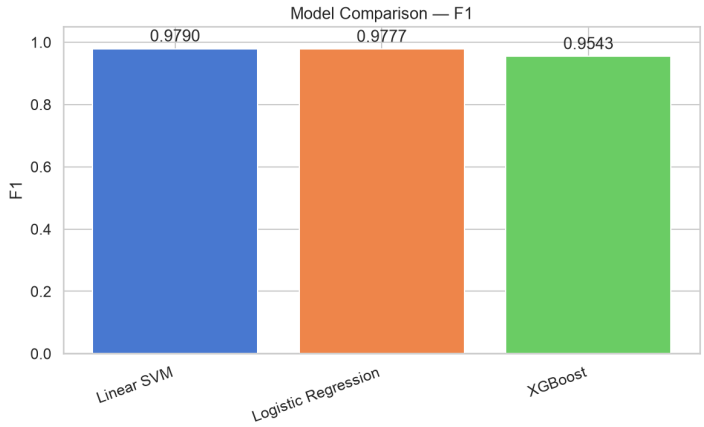
Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
Linear SVM	98.61%	98.31%	97.49%	0.9790	0.9986	0.9976	0.9686
Logistic Regression	98.53%	97.98%	97.56%	0.9777	0.9987	0.9977	0.9668
XGBoost	96.95%	94.52%	96.37%	0.9543	0.9943	0.9876	0.9315

Main reading: Linear SVM has the highest observed accuracy and F1-score, while Logistic Regression is very close. The paired significance test below does not establish a statistically significant difference between those two models.

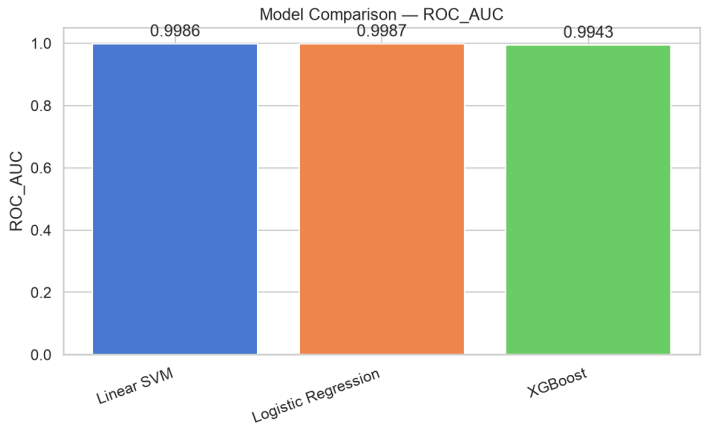
Accuracy comparison



F1-score comparison



ROC-AUC comparison



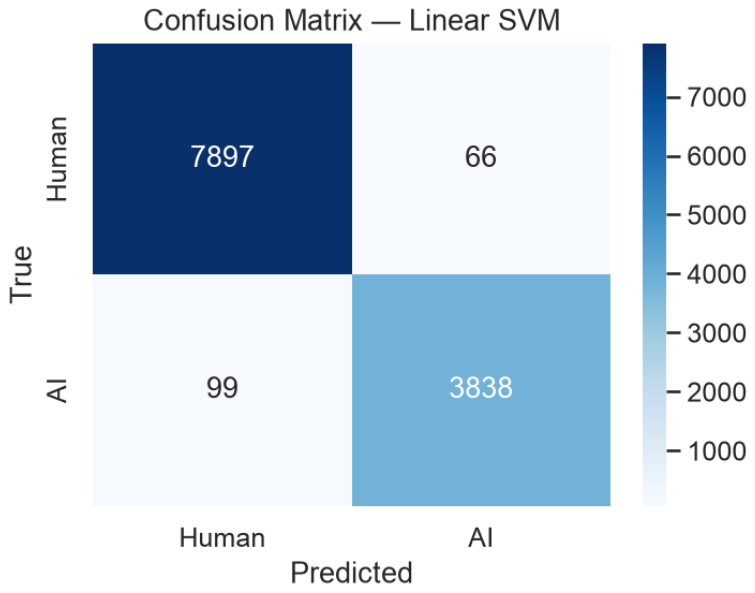
2. Confusion matrices and paired statistical checks

Confusion-matrix counts use Human-Written as the negative class and AI-Generated as the positive class.

Model	TN	FP	FN	TP	Total
Linear SVM	7,897	66	99	3,838	11,900
Logistic Regression	7,884	79	96	3,841	11,900
XGBoost	7,743	220	143	3,794	11,900

Model comparison	Both correct	A only correct	B only correct	Both wrong	Corrected χ^2	p-value	Conclusion
Linear SVM vs Logistic Regression	11,711	24	14	151	2.1316	0.1443	Not significant
Linear SVM vs XGBoost	11,389	346	148	17	78.5607	<0.000001	Significant
Logistic Regression vs XGBoost	11,380	345	157	18	69.6594	<0.000001	Significant

Interpretation: SVM and Logistic Regression are not significantly different under the paired McNemar test ($p = 0.1443$). Both linear models differ significantly from XGBoost on this held-out partition ($p < 0.000001$). These results apply to this benchmark and split; they do not prove universal authorship detection.

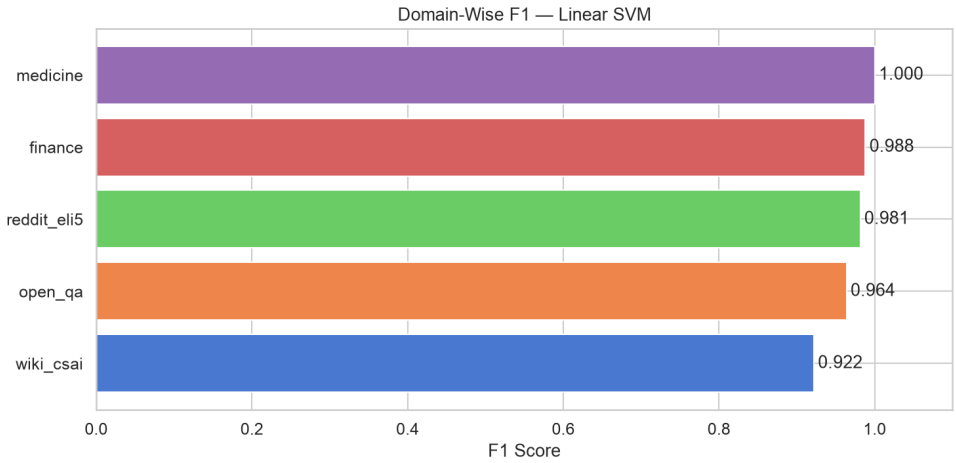


3. Domain-level results and explainability

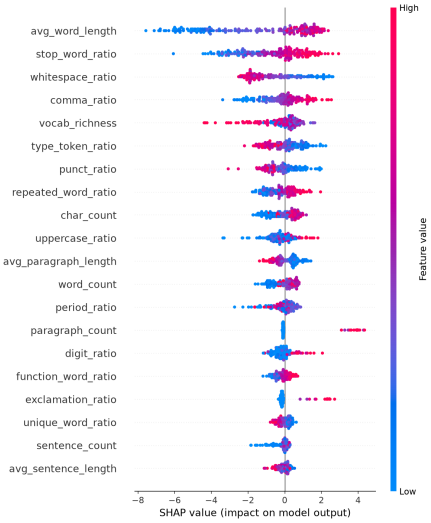
Domain	Test samples	LR F1	SVM F1	XGBoost F1
medicine	385	0.9975	1.0000	0.9757
finance	1,260	0.9881	0.9880	0.9164
reddit_eli5	9,264	0.9798	0.9814	0.9704
open_qa	763	0.9620	0.9638	0.9511
wiki_csai	228	0.9261	0.9219	0.8611

The domain results are based on test-set samples, not the full domain totals. Performance is weakest in wiki_csai for these models, which reinforces the need for domain-specific evaluation.

Linear SVM F1 by domain



SHAP summary for XGBoost



Top saved XGBoost stylometric features include average word length (22.44%), paragraph count (11.67%), unique-word ratio (11.52%), type-token ratio (6.40%), and character count (5.44%). Feature importance describes behaviour of this trained model; it is not a universal rule for detecting AI authorship.

4. Colab runtime and demonstration outputs

The accompanying Colab notebook is the execution step. It uploads this code package, installs the lightweight runtime requirements, loads the three saved models, runs the automated tests and statistical verification, executes demonstration predictions, and regenerates this PDF with the actual runtime record. The current packaged PDF is not a substitute for running that notebook.

Suggested Colab workflow

1. Open the notebook with Google Colab and run cells from top to bottom.
2. When prompted, upload I26043592_Vishnu_Vardhan_Code.zip from your Downloads folder.
3. Review the test summary, independent statistical verification, three-model inference table, and document-parser checks.
4. Download the regenerated outputs PDF and updated ZIP from the notebook final cell.

Interpretation and limitations

The classifier estimates whether text resembles the HC3 human-written or ChatGPT-generated examples. High benchmark scores do not prove authorship or guarantee performance on different generators, domains, editing styles, or very short text. Predictions should be treated as a research aid and interpreted with human review.

Source artefacts: results/metrics/model_comparison.csv; statistical_tests_verified.csv; confusion_matrices_verified.csv; results/error_analysis/*_domain_metrics.csv; figures under results/figures/. All values in this report should be kept aligned with the executable project.